

Mathematics Refresher

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SAP SE / DHBW CAS Heilbronn

Summer term 2026



Find all slides on [GitHub](#) (DaWe1992/Applied_ML_Fundamentals)

Agenda for this Unit

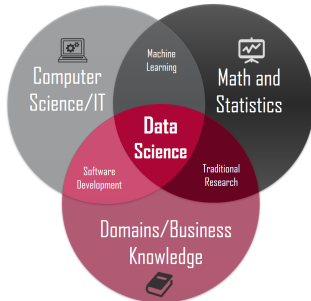
- 1 Introduction
- 2 Linear Algebra
- 3 Probability Theory and Statistics
- 4 Wrap-Up

Section: Introduction

Introduction
Math is important!

Introduction

Mathematics play a major role in data science and machine learning!

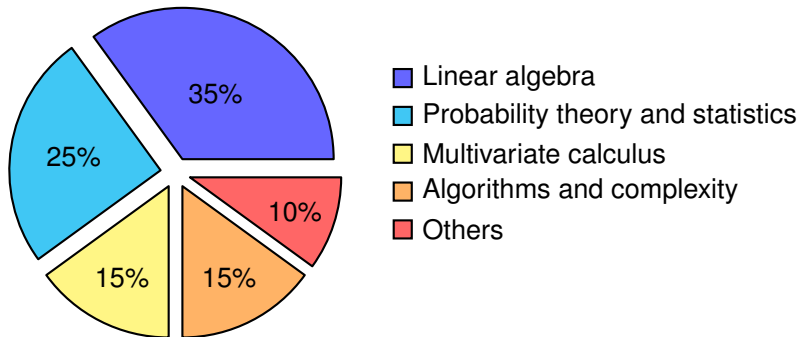


You will need it to understand:

- **Statistical** machine learning
- How **optimization** is used in learning and empirical risk minimization
- How linear algebra, calculus, and statistics are used to make learning and inference more efficient

Math is important!

Rough importance of mathematical disciplines in data science and machine learning:



Section: Linear Algebra

Vectors and Vector Operations
Matrix Operations
Determinants and Inverses
Eigenvectors and Eigenvalues
Symmetric Matrices and Definiteness

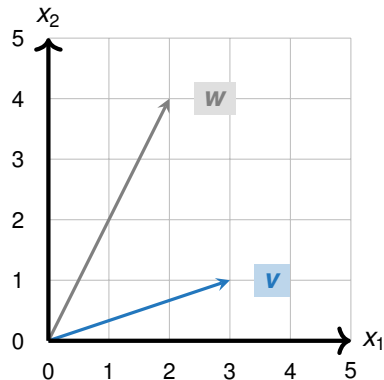
What is a Vector?

General:

$$\mathbf{x} = \begin{pmatrix} x_1 \\ \vdots \\ x_D \end{pmatrix} \in \mathbb{R}^D$$

Example: ($D = 2$)

$$\mathbf{v} = \begin{pmatrix} 3 \\ 1 \end{pmatrix} \quad \mathbf{w} = \begin{pmatrix} 2 \\ 4 \end{pmatrix}$$



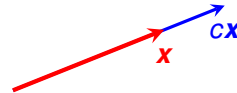
Multiplication of Vectors by Scalars

General: Let $c \in \mathbb{R}$ and $\mathbf{x} \in \mathbb{R}^D$:

$$c\mathbf{x} = c \begin{pmatrix} x_1 \\ \vdots \\ x_D \end{pmatrix} = \begin{pmatrix} cx_1 \\ \vdots \\ cx_D \end{pmatrix}$$

Example: ($D = 2$)

$$2\mathbf{v} = 2 \begin{pmatrix} 3 \\ 1 \end{pmatrix} = \begin{pmatrix} 6 \\ 2 \end{pmatrix}$$



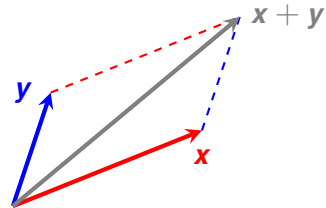
Addition of Vectors

General: Let $\mathbf{x}, \mathbf{y} \in \mathbb{R}^D$

$$\mathbf{x} + \mathbf{y} = \begin{pmatrix} x_1 \\ \vdots \\ x_D \end{pmatrix} + \begin{pmatrix} y_1 \\ \vdots \\ y_D \end{pmatrix} = \begin{pmatrix} x_1 + y_1 \\ \vdots \\ x_D + y_D \end{pmatrix}$$

Example: ($D = 2$)

$$\mathbf{v} + \mathbf{w} = \begin{pmatrix} 3 \\ 1 \end{pmatrix} + \begin{pmatrix} 2 \\ 4 \end{pmatrix} = \begin{pmatrix} 5 \\ 5 \end{pmatrix}$$



Linear Combination of Vectors and Span

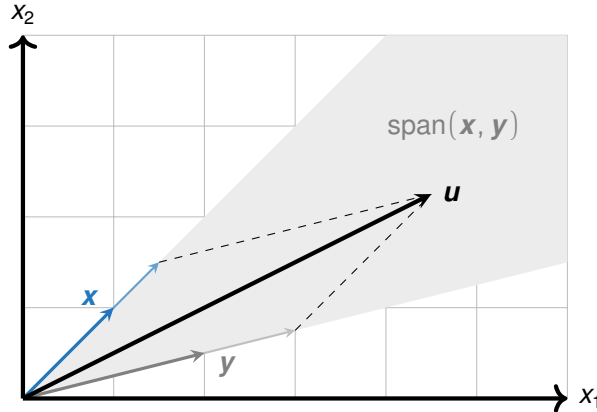
- Let $c_1, \dots, c_N \in \mathbb{R}$ and $\mathbf{x}^1, \dots, \mathbf{x}^N \in \mathbb{R}^D$
- A **linear combination** of these vectors is given by $\mathbf{u} \in \mathbb{R}^D$:

$$\mathbf{u} := \sum_{n=1}^N c_n \mathbf{x}^n \quad (1)$$

- The **span** (German: *lineare Hülle*) of $\mathbf{x}^1, \dots, \mathbf{x}^N \in \mathbb{R}^D$ is defined by:

$$\text{span}(\mathbf{x}^1, \dots, \mathbf{x}^N) := \left\{ \mathbf{u} \in \mathbb{R}^D : \exists c_1, \dots, c_N : \mathbf{u} = \sum_{n=1}^N c_n \mathbf{x}^n \right\} \quad (2)$$

Linear Combination of Vectors and Span (Ctd.)



Vector Transpose, Inner and Outer Product

- Let $\mathbf{x}, \mathbf{y} \in \mathbb{R}^D$ be given
- Transposition:**

$$\mathbf{x} = \begin{pmatrix} x_1 \\ \vdots \\ x_D \end{pmatrix} \quad \mathbf{x}^\top = \begin{pmatrix} x_1 & \dots & x_D \end{pmatrix} \quad (3)$$

- Inner product** (also referred to as **dot product** or **scalar product**):

$$\mathbf{x}^\top \mathbf{y} \equiv \langle \mathbf{x}, \mathbf{y} \rangle := \begin{pmatrix} x_1 & \dots & x_D \end{pmatrix} \begin{pmatrix} y_1 \\ \vdots \\ y_D \end{pmatrix} = \sum_{d=1}^D x_d y_d \quad (4)$$



Vector Transpose, Inner and Outer Product (Ctd.)

- Outer product:

$$\mathbf{xy}^T = \begin{pmatrix} x_1 \\ \vdots \\ x_D \end{pmatrix} \begin{pmatrix} y_1 & \dots & y_D \end{pmatrix} = \begin{pmatrix} x_1 y_1 & x_1 y_2 & \dots & x_1 y_D \\ x_2 y_1 & x_2 y_2 & \dots & x_2 y_D \\ \vdots & \vdots & \ddots & \vdots \\ x_D y_1 & x_D y_2 & \dots & x_D y_D \end{pmatrix} \quad (5)$$

Remember: The inner product yields a scalar value; The result of an outer product is a matrix!

Example: Vector Transpose, Inner and Outer Product

- Let $\mathbf{v} = \begin{pmatrix} 3 \\ 1 \end{pmatrix} \in \mathbb{R}^2$ and $\mathbf{w} = \begin{pmatrix} 2 \\ 4 \end{pmatrix} \in \mathbb{R}^2$

- Transposition:**

$$\mathbf{v}^T = \begin{pmatrix} 3 & 1 \end{pmatrix}$$

- Inner product:**

$$\mathbf{v}^T \mathbf{w} = \begin{pmatrix} 3 & 1 \end{pmatrix} \begin{pmatrix} 2 \\ 4 \end{pmatrix} = 3 \cdot 2 + 1 \cdot 4 = 10$$

Example: Vector Transpose, Inner and Outer Product (Ctd.)

- **Outer product:**

$$\mathbf{vw}^T = \begin{pmatrix} 3 \\ 1 \end{pmatrix} \begin{pmatrix} 2 & 4 \end{pmatrix} = \begin{pmatrix} 6 & 12 \\ 2 & 4 \end{pmatrix}$$

Length of a Vector and Norms

- Length of a vector (**EUCLIDEAN norm**): Let $\mathbf{x}, \mathbf{y} \in \mathbb{R}^D$ and $c \in \mathbb{R}$

$$\|\mathbf{x}\| := \sqrt{\mathbf{x}^\top \mathbf{x}} \quad (6)$$

$$\|c\mathbf{x}\| = |c| \cdot \|\mathbf{x}\| \quad (7)$$

$$\|\mathbf{x} + \mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\| \quad (8)$$

- Example: Let $\mathbf{v} = \begin{pmatrix} 3 \\ 1 \end{pmatrix} \in \mathbb{R}^2$

$$\|\mathbf{v}\| = \sqrt{3^2 + 1^2} = \sqrt{10}$$

Angle between Vectors

- The **angle** between two vectors $\mathbf{x}, \mathbf{y} \in \mathbb{R}^D$ is given by:

$$\cos \angle(\mathbf{x}, \mathbf{y}) = \frac{\mathbf{x}^\top \mathbf{y}}{\|\mathbf{x}\| \cdot \|\mathbf{y}\|} = \frac{\sum_{d=1}^D x_d y_d}{\sqrt{\sum_{d=1}^D x_d^2} \cdot \sqrt{\sum_{d=1}^D y_d^2}} \quad (9)$$

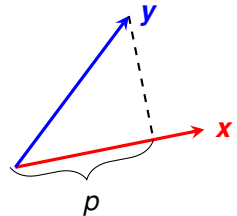
$$\cos \angle(\mathbf{v}, \mathbf{w}) = \frac{\mathbf{v}^\top \mathbf{w}}{\|\mathbf{v}\| \cdot \|\mathbf{w}\|} = \frac{10}{\sqrt{10} \cdot \sqrt{20}} \approx 0.71$$

- Inner product: $\mathbf{x}^\top \mathbf{y} = \|\mathbf{x}\| \cdot \|\mathbf{y}\| \cdot \cos \angle(\mathbf{x}, \mathbf{y})$

(Orthogonal) Projection of Vectors

- How is the projection of $\mathbf{y}$ onto $\mathbf{x}$ defined?
- Formally, we have:

$$\begin{aligned} p &= \|\mathbf{y}\| \cos \angle(\mathbf{x}, \mathbf{y}) \\ &= \|\mathbf{y}\| \frac{\mathbf{x}^\top \mathbf{y}}{\|\mathbf{x}\| \cdot \|\mathbf{y}\|} \\ &= \frac{\mathbf{x}^\top \mathbf{y}}{\|\mathbf{x}\|} \end{aligned} \tag{10}$$



- Note that p is **not a vector**!

What is a Matrix?

- **General case:**

$$\mathbf{X} = \begin{pmatrix} x_{11} & x_{12} & \dots & x_{1M} \\ x_{21} & x_{22} & \dots & x_{2M} \\ \vdots & \vdots & \ddots & \vdots \\ x_{N1} & x_{N2} & \dots & x_{NM} \end{pmatrix} \in \mathbb{R}^{N \times M}$$

- x_{nm} is the entry in row n and column m

Remember: Zeilen zuerst, Spalten später (German)

Example: Matrices

$$\mathbf{A} = \begin{pmatrix} 3 & 4 & 5 \\ 1 & 0 & 1 \end{pmatrix} \in \mathbb{R}^{2 \times 3}$$

$\mathbf{A}$ has three rows, but only two columns.

$$\mathbf{B} = \begin{pmatrix} 10 & 1 \\ 11 & 2 \end{pmatrix} \in \mathbb{R}^{2 \times 2}$$

$\mathbf{B}$ is a **square matrix** as it has the same number of rows and columns.

Square matrices play a special role in mathematics, e.g. matrix inversion and determinants are only defined for square matrices.

$$\mathbf{C} = \begin{pmatrix} 3 & 0 \\ 0 & 7 \end{pmatrix} \in \mathbb{R}^{2 \times 2}$$

$\mathbf{C}$ is a square matrix, but also a **diagonal matrix** because $c_{ij} = 0$ for $i \neq j$. We often write $\mathbf{C} := \text{diag}(3, 7)$.

Special Matrices

Identity matrix:

$$\mathbf{I}_N := \begin{pmatrix} 1 & 0 & 0 & \dots & 0 \\ 0 & 1 & 0 & \dots & 0 \\ 0 & 0 & 1 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 1 \end{pmatrix} \in \mathbb{R}^{N \times N}$$

Zero matrix:

$$\mathbf{0} := \begin{pmatrix} 0 & 0 & 0 & \dots & 0 \\ 0 & 0 & 0 & \dots & 0 \\ 0 & 0 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 0 \end{pmatrix} \in \mathbb{R}^{N \times N}$$



Matrix Transpose

- **Matrix transposition:**

$$\mathbf{X}^T = \begin{pmatrix} x_{11} & x_{21} & \dots & x_{N1} \\ x_{12} & x_{22} & \dots & x_{N2} \\ \vdots & \vdots & \ddots & \vdots \\ x_{1M} & x_{2M} & \dots & x_{NM} \end{pmatrix} \in \mathbb{R}^{M \times N} \quad (11)$$

- **Please note:** $\mathbf{X} \in \mathbb{R}^{N \times M}$, but $\mathbf{X}^T \in \mathbb{R}^{M \times N}$



Matrix Addition

- **Addition of matrices:** Let $\mathbf{X}, \mathbf{Y} \in \mathbb{R}^{N \times M}$

$$\begin{aligned}\mathbf{X} + \mathbf{Y} &= \begin{pmatrix} x_{11} & \dots & x_{1M} \\ \vdots & \ddots & \vdots \\ x_{N1} & \dots & x_{NM} \end{pmatrix} + \begin{pmatrix} y_{11} & \dots & y_{1M} \\ \vdots & \ddots & \vdots \\ y_{N1} & \dots & y_{NM} \end{pmatrix} \\ &= \begin{pmatrix} x_{11} + y_{11} & \dots & x_{1M} + y_{1M} \\ \vdots & \ddots & \vdots \\ x_{N1} + y_{N1} & \dots & x_{NM} + y_{NM} \end{pmatrix} \in \mathbb{R}^{N \times M} \quad (12)\end{aligned}$$

- **Please note:** $\mathbf{X}$ and $\mathbf{Y}$ must be of the same size!



Multiplication of Matrices by Scalars

- **Multiplication by scalars:** Let $\mathbf{X} \in \mathbb{R}^{N \times M}$ and $c \in \mathbb{R}$

$$c\mathbf{X} = c \begin{pmatrix} x_{11} & x_{12} & \dots & x_{1M} \\ x_{21} & x_{22} & \dots & x_{2M} \\ \vdots & \vdots & \ddots & \vdots \\ x_{N1} & x_{N2} & \dots & x_{NM} \end{pmatrix} = \begin{pmatrix} cx_{11} & cx_{12} & \dots & cx_{1M} \\ cx_{21} & cx_{22} & \dots & cx_{2M} \\ \vdots & \vdots & \ddots & \vdots \\ cx_{N1} & cx_{N2} & \dots & cx_{NM} \end{pmatrix} \in \mathbb{R}^{N \times M} \quad (13)$$

- This is defined for all matrices



Multiplication of Matrices by Vectors

- **Matrix-vector multiplication:** Let $\mathbf{X} \in \mathbb{R}^{N \times M}$ and $\mathbf{y} \in \mathbb{R}^M$

$$\mathbf{X}\mathbf{y} = \begin{pmatrix} x_{11} & x_{12} & \dots & x_{1M} \\ x_{21} & x_{22} & \dots & x_{2M} \\ \vdots & \vdots & \ddots & \vdots \\ x_{N1} & x_{N2} & \dots & x_{NM} \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_M \end{pmatrix} = \begin{pmatrix} \sum_{m=1}^M x_{1m}y_m \\ \sum_{m=1}^M x_{2m}y_m \\ \vdots \\ \sum_{m=1}^M x_{Nm}y_m \end{pmatrix} \in \mathbb{R}^N \quad (14)$$

- **Please note:** The number of columns of $\mathbf{X}$ and the number of rows of $\mathbf{y}$ must be equal in order for the matrix-vector product to exist!
- The order is important: $\mathbf{X}\mathbf{y}$ is defined, but $\mathbf{y}\mathbf{X}$ is not, if $n > 1$



Matrix Multiplication (Ctd.)

- **Matrix-matrix multiplication:** Let $\mathbf{X} \in \mathbb{R}^{L \times M}$ and $\mathbf{Y} \in \mathbb{R}^{M \times N}$

$$\mathbf{XY} = \begin{pmatrix} x_{11} & \dots & x_{1M} \\ \vdots & \ddots & \vdots \\ x_{L1} & \dots & x_{LM} \end{pmatrix} \begin{pmatrix} y_{11} & \dots & y_{1N} \\ \vdots & \ddots & \vdots \\ y_{M1} & \dots & y_{MN} \end{pmatrix} = \begin{pmatrix} z_{11} & \dots & z_{1N} \\ \vdots & \ddots & \vdots \\ z_{L1} & \dots & z_{LN} \end{pmatrix} \quad (15)$$

where:

$$z_{\ell n} = \sum_{m=1}^M x_{\ell m} y_{m n} \quad (16)$$

- **Please note:** The number of columns of $\mathbf{X}$ and the number of rows of $\mathbf{Y}$ must match!

Determinants of Square Matrices

- **Determinants are defined for square matrices only!**
- The **determinant** of a (2×2) -matrix is given by:

$$\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} := ad - bc \quad (17)$$

- The determinant of a (3×3) -matrix is given by **(rule of SARRUS)**:

$$\det \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} := aei + bfg + cdh - gec - hfa - idb \quad (18)$$

LAPLACE Expansion

Use the **LAPLACE expansion** for $(N \times N)$ -matrices if $N > 3$:

LAPLACE expansion: Let $\mathbf{X} \in \mathbb{R}^{N \times N}$ be given. Then

$$\det(\mathbf{X}) = \sum_{n=1}^N x_{nm} \cdot (-1)^{n+m} \cdot \det(\mathbf{X}_{nm}), \quad (19)$$

where $\mathbf{X}_{nm}$ is the matrix obtained by removing row n and column m from $\mathbf{X}$.

Matrix Inversion

- **Matrix inversion is defined for square matrices only!**
- $\mathbf{X} \in \mathbb{R}^{N \times N}$ multiplied by its inverse $\mathbf{X}^{-1} \in \mathbb{R}^{N \times N}$ gives the identity matrix $\mathbf{I}_N$:

$$\mathbf{X}\mathbf{X}^{-1} = \mathbf{I}_N \quad (20)$$

- Also, the order is not important, i. e.:

$$\mathbf{X}^{-1}\mathbf{X} = \mathbf{I}_N \quad (21)$$

- We call $\mathbf{X}$ **non-singular** or **invertible**, if $\mathbf{X}^{-1}$ exists

Matrix Inversion (Ctd.)

Let $\mathbf{X} \in \mathbb{R}^{N \times N}$ be a square matrix. The following statements are equivalent:

$\mathbf{X}$ is invertible $\iff \mathbf{X}$ is non-singular

$\iff \det(\mathbf{X}) \neq 0$

$\iff \mathbf{X}$ has rank N (full rank)

$\iff \mathbf{X}$ does not have eigenvalue 0

$\iff$ The **reduced row echelon form** of $\mathbf{X}$ is the identity matrix $\mathbf{I}_N$

Matrix Inversion (Ctd.)

- The inverse of a matrix can be computed using the **GAUSS-JORDAN algorithm**
- Special case:** Do not use the GAUSS-JORDAN algorithm for (2×2) -matrices!

You can be more efficient using the identity

$$\mathbf{X} \operatorname{adj}(\mathbf{X}) = \det(\mathbf{X}) \mathbf{I}_N \quad \stackrel{\det(\mathbf{X}) \neq 0}{\iff} \quad \mathbf{X} \overbrace{\frac{1}{\det(\mathbf{X})} \operatorname{adj}(\mathbf{X})}^{=: \mathbf{X}^{-1}} = \mathbf{I}_N, \quad (22)$$

where $\operatorname{adj}(\mathbf{X})$ is the **adjugate matrix** of $\mathbf{X}$.

Matrix Inversion (Ctd.)

- In general, $\text{adj}(\mathbf{X})$ is hard to compute, but it is easy for (2×2) -matrices
- Let

$$\mathbf{X} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathbb{R}^{2 \times 2}$$

- The adjugate matrix $\text{adj}(\mathbf{X})$ of $\mathbf{X}$ is then given by:

$$\text{adj}(\mathbf{X}) := \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} \in \mathbb{R}^{2 \times 2} \quad (23)$$



Eigenvectors and Eigenvalues

- Let $\mathbf{X} \in \mathbb{R}^{N \times N}$ be a square matrix
- Some vectors $\mathbf{v} \in \mathbb{R}^N$ only change their length (but not their direction) when multiplied by $\mathbf{X}$
- Such vectors are called **eigenvectors** of $\mathbf{X}$ and the scaling factors are known as **eigenvalues** of $\mathbf{X}$

Eigenvectors and eigenvalues satisfy the equation:

$$\mathbf{X}\mathbf{v} = \lambda\mathbf{v} \tag{24}$$

Example: Eigenvectors and Eigenvalues

- Let $\mathbf{X} := \begin{pmatrix} 4 & -1 \\ 2 & 1 \end{pmatrix} \in \mathbb{R}^{2 \times 2}$ be given

- We have (*please verify for yourself*):

$$\begin{pmatrix} 4 & -1 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 2 \\ 4 \end{pmatrix} = 2 \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

- Thus, $\mathbf{v} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$ is an eigenvector of $\mathbf{X}$ and $\lambda = 2$ is the corresponding eigenvalue

Eigenvectors form a Basis (Eigenbasis)

- Let $\mathbf{v}^1, \mathbf{v}^2, \dots, \mathbf{v}^N$ be N eigenvectors of $\mathbf{X} \in \mathbb{R}^{N \times N}$ with corresponding eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_N$
- **Theorem (*)**:
 - If $\mathbf{v}^1, \mathbf{v}^2, \dots, \mathbf{v}^N$ correspond to **distinct eigenvalues** $\lambda_1, \lambda_2, \dots, \lambda_N$, then the system $(\mathbf{v}^1, \mathbf{v}^2, \dots, \mathbf{v}^N)$ represents a basis of $\mathbb{R}^N$ (**eigenbasis**)
 - Hence, $\mathbf{v}^1, \dots, \mathbf{v}^N$ are **linearly independent** and any vector $\mathbf{u} \in \mathbb{R}^N$ can be **uniquely** expressed as a **linear combination** of the eigenvectors of $\mathbf{X}$, i. e.
 $\exists c_1, \dots, c_N \in \mathbb{R}$ such that

$$\mathbf{u} = \sum_{n=1}^N c_n \mathbf{v}^n \quad \forall \mathbf{u} \in \mathbb{R}^N$$

How to compute Eigenvalues and Eigenvectors

- Let $\mathbf{X} \in \mathbb{R}^{N \times N}$ be a square matrix
- The eigenvalues are the roots (German: *Nullstellen*) of the **characteristic polynomial** defined by

$$\chi_{\mathbf{X}}(\lambda) := \det(\lambda \mathbf{I}_N - \mathbf{X}) \quad (25)$$

- For each eigenvalue λ_n we have to solve the **homogeneous system of linear equations**

$$(\mathbf{X} - \lambda_n \mathbf{I}_N) \mathbf{v} = \mathbf{0} \quad (26)$$

to obtain the respective eigenvectors (see $\Rightarrow$ [here](#))

Example: Computation of Eigenvalues and Eigenvectors

Computation of the eigenvalues:

- Let $\mathbf{A} := \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix} \in \mathbb{R}^{2 \times 2}$

- The eigenvalues are the roots of (25) which is given by

$$\chi_{\mathbf{A}} := \det \begin{pmatrix} \lambda - 2 & -1 \\ -1 & \lambda - 2 \end{pmatrix} = (\lambda - 2)^2 - 1 = (\lambda - 1)(\lambda - 3)$$

- We directly see: $\lambda_1 = 1, \lambda_2 = 3$

Example: Computation of Eigenvalues and Eigenvectors (Ctd.)

Computation of the eigenspaces:

- We start with the eigenvalue $\lambda_1 = 1$ and solve (26):

$$(\mathbf{A} - \mathbf{I}_2) = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \xrightarrow{I+II} \begin{pmatrix} 1 & -1 \\ 0 & 0 \end{pmatrix}$$

- The eigenspace for eigenvalue 1 is therefore given by

$$\mathcal{E}(1) = \{t \cdot (-1, -1)^\top : t \in \mathbb{R}\}$$

- Similarly, we can show that $\mathcal{E}(3) = \{t \cdot (1, -1)^\top : t \in \mathbb{R}\}$

Diagonalizable Matrices

- Let $\mathbf{A} \in \mathbb{R}^{N \times N}$ be a square matrix
- If the conditions of theorem (*) are met, then we can find a non-singular matrix $\mathbf{S} \in \mathbb{R}^{N \times N}$ such that:

$$\mathbf{S}^{-1} \mathbf{A} \mathbf{S} = \mathbf{B} \in \mathbb{R}^{N \times N} \quad (27)$$

- The columns of $\mathbf{S}$ are given by the eigenvectors of $\mathbf{A}$, and $\mathbf{B} := \text{diag}(\lambda_1, \dots, \lambda_N)$ is a diagonal matrix containing the eigenvalues of $\mathbf{A}$
- We say $\mathbf{A}$ is a **diagonalizable matrix**
- $\mathbf{A}$ and $\mathbf{B}$ are called **similar matrices**



Symmetric Matrices

- A square $(N \times N)$ -matrix $\mathbf{X}$ is called **symmetric**, if and only if

$$\mathbf{X} = \mathbf{X}^\top \quad (28)$$

- **Some properties:**
 - The inverse $\mathbf{X}^{-1}$ of $\mathbf{X}$ is also a symmetric matrix
 - **Eigen-decomposition:** Let $\mathbf{X}$ be a symmetric matrix. In this case the conditions of theorem (*) are met and we can find an **orthogonal matrix** $\mathbf{Q}$ (i. e. $\mathbf{Q}^{-1} = \mathbf{Q}^\top$) such that $\mathbf{Q}^\top \mathbf{X} \mathbf{Q} = \mathbf{D}$. The columns of $\mathbf{Q}$ are given by the normalized eigenvectors of $\mathbf{X}$, and $\mathbf{D}$ is a diagonal matrix whose entries are the corresponding eigenvalues

Example: Eigen-Decomposition

- Consider $\mathbf{A} := \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix} \in \mathbb{R}^{2 \times 2}$
- We choose one eigenvector for each eigenvalue and divide them by their lengths:

$$\text{Eigenvalue 1: } \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right)^\top \quad \text{Eigenvalue 3: } \left(\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}} \right)^\top$$

Thus, the eigen-decomposition of $\mathbf{A}$ is given by:

$$\mathbf{D} = \begin{pmatrix} 1 & 0 \\ 0 & 3 \end{pmatrix} = \begin{pmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & -1/\sqrt{2} \end{pmatrix} \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix} \begin{pmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & -1/\sqrt{2} \end{pmatrix}^\top = \mathbf{Q}^\top \mathbf{A} \mathbf{Q}$$

Positive (semi-)definite Matrices

- A symmetric matrix $\mathbf{X} \in \mathbb{R}^{N \times N}$ is called **positive definite** (notation: $\mathbf{X} \succ 0$), if

$$\mathbf{z}^T \mathbf{X} \mathbf{z} > 0 \quad \forall \mathbf{z} \in \mathbb{R}^N \setminus \{\mathbf{0}\} \quad (29)$$

- Or **positive semi-definite** (notation: $\mathbf{X} \succeq 0$), if

$$\mathbf{z}^T \mathbf{X} \mathbf{z} \geq 0 \quad \forall \mathbf{z} \in \mathbb{R}^N \quad (30)$$

Such matrices are important in machine learning. For instance, the covariance matrices Σ are always positive semi-definite.

Section:

Probability Theory and Statistics

Random Variables and Common Distributions
Basic Rules of Probability
Expectation and Variance
Kullback-Leibler Divergence

Random Variables

- What is a **random variable**?

Random Variables

- What is a **random variable**?
 - It's a random number determined by chance (according to an underlying distribution).
To be precise: A random variable $\mathcal{X}$ is a **measurable function**

$$\mathcal{X} : \Omega \rightarrow \mathbb{R} \quad \text{where } \Omega \text{ is the } \mathbf{sample\ space}$$

- Examples of random variables in machine learning: Input data, output data, noise
- What is a **probability distribution**?

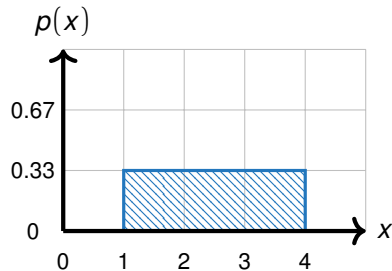
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- Examples of random variables in machine learning: Input data, output data, noise
- What is a **probability distribution**?
 - It describes the probability that a random variable is equal to a certain value
 - It can be given by the physics of an experiment (e. g. throwing dice)
 - **Discrete** vs. **continuous** distributions

Uniform Distribution



Every outcome is equally probable within a bounded region $\mathcal{R} := [a, b]$

$$p(\mathcal{X} = x) := \frac{1}{b - a}$$

Discrete Distributions

A **discrete random variable** takes on discrete values.

Please note: Discrete does not mean finite!

Examples:

- When throwing a die, the possible values are given by the finite set:

$$\mathcal{X} \in \{1, 2, 3, 4, 5, 6\}$$

- The number of sand grains at the beach (countably infinite set):

$$\mathcal{X} \in \mathbb{N}$$

Discrete Distributions (Ctd.)

- All probabilities sum up to 1, i. e.:

$$\sum_{x \in \mathcal{X}(\Omega)} p(\mathcal{X} = x) = 1$$

- Discrete distributions are particularly important in classification
- A discrete distribution is described by a **probability mass function** (also called **frequency function**)

BERNOULLI Distribution

- A **BERNOULLI random variable** only takes on two values (e. g. 0 and 1):

$$\mathcal{X} \in \{0, 1\} \quad (31)$$

$$p(\mathcal{X} = 1; \mu) = \mu \quad (32)$$

$$p(\mathcal{X} = 0; \mu) = 1 - \mu \quad (33)$$

$$\mathbb{E}\{\mathcal{X}\} = \mu \quad (34)$$

$$\mathbb{V}\{\mathcal{X}\} = \mu(1 - \mu) \quad (35)$$

- The BERNOLLI distribution is governed only by the parameter μ , the **probability of success**



Binomial Distribution

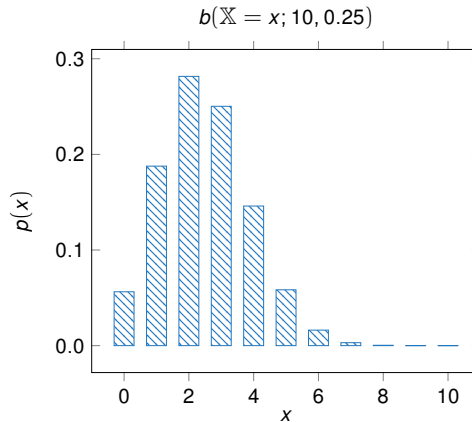
- Repeating a BERNOULLI experiment N times leads to the **binomial distribution**
- **Example:** What is the probability of getting $n \in \mathbb{N}$ heads in N trials?

$$b(\mathcal{X} = n; N, \mu) := \binom{N}{n} \mu^n (1 - \mu)^{N-n} \quad (36)$$

$$\mathbb{E}\{\mathcal{X}\} = N\mu \quad (37)$$

$$\mathbb{V}\{\mathcal{X}\} = N\mu(1 - \mu) \quad (38)$$

Binomial Distribution (Ctd.)



Continuous Distributions

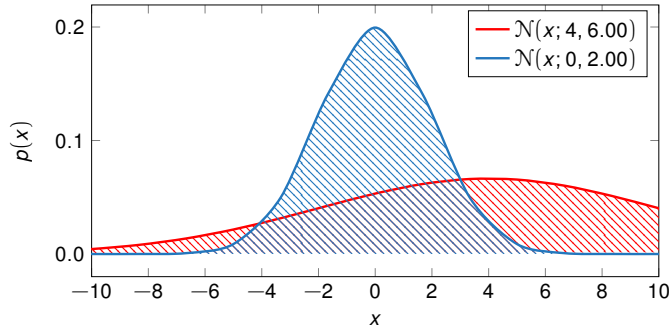
Continuous random variables take on continuous values

- Continuous distributions are discrete distributions where the **number of discrete values goes to infinity**, while the **probability of each value goes to zero**
- A continuous random variable $\mathcal{X}$ is described by a **probability density function** which integrates to 1, i. e.:

$$\int_{-\infty}^{\infty} p(\mathcal{X} = x) dx = 1$$



Gaussian Distribution



$$\mathcal{N}(\mathcal{X} = x; \mu, \sigma^2) := \frac{1}{\sqrt{2\pi\sigma^2}} \cdot \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right) \quad (39)$$



Central Limit Theorem

Central Limit Theorem:

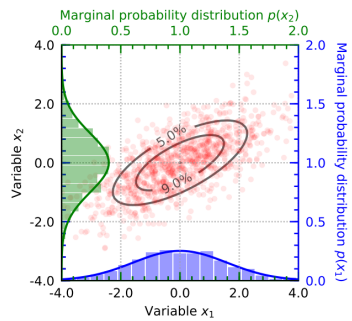
The distribution of the sum of N i.i.d. (independent and identically distributed) random variables **becomes increasingly GAUSSIAN as N increases**

- The GAUSSIAN distribution is one of the most important distributions
- GAUSSIAN distributions often are a good model (due to the central limit theorem)
- Working with GAUSSIANS leads to **analytical solutions for complex operations**



Multivariate GAUSSian Distribution

$$\mathcal{N}_D(\mathbf{x}; \boldsymbol{\mu}, \boldsymbol{\Sigma}) := \frac{1}{\sqrt{(2\pi)^D \det(\boldsymbol{\Sigma})}} \cdot \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu})\right) \quad (40)$$



Please note: $\mathbf{x}$ and $\boldsymbol{\mu}$ are vectors, while $\boldsymbol{\Sigma}$ is a matrix.

The probability given by $\mathcal{N}_D(\mathbf{x}; \boldsymbol{\mu}, \boldsymbol{\Sigma})$ is still a scalar value!

Basic Rules of Probability

- **Joint distribution:**

$$p(\mathcal{X} \cap \mathcal{Y}) \quad (41)$$

- **Marginal distribution:**

$$p(\mathcal{Y}) = \int_{\mathcal{X}} p(\mathcal{X} \cap \mathcal{Y}) \, d\mathcal{X} \quad (42)$$

- **Conditional distribution:**

$$p(\mathcal{Y}|\mathcal{X}) = \frac{p(\mathcal{X} \cap \mathcal{Y})}{p(\mathcal{X})} \quad (43)$$

Basic Rules of Probability (Ctd.)

- **Probabilistic independence:**

$$p(\mathcal{X} \cap \mathcal{Y}) = p(\mathcal{X})p(\mathcal{Y}) \quad (44)$$

- **Chain rule of probabilities:**

$$\begin{aligned} p(\mathcal{X}_1 \cap \dots \cap \mathcal{X}_N) &= p(\mathcal{X}_1 | \mathcal{X}_2 \cap \dots \cap \mathcal{X}_N) p(\mathcal{X}_2 \cap \dots \cap \mathcal{X}_N) \\ &= p(\mathcal{X}_1 | \mathcal{X}_2 \cap \dots \cap \mathcal{X}_N) \dots p(\mathcal{X}_{N-1} | \mathcal{X}_N) p(\mathcal{X}_N) \end{aligned} \quad (45)$$

- **BAYES' rule:**

$$p(\mathcal{Y} | \mathcal{X}) = \frac{p(\mathcal{X} | \mathcal{Y}) p(\mathcal{Y})}{p(\mathcal{X})} \quad (46)$$



Expectation

- The **expectation** $\mathbb{E}$ of a random variable $\mathcal{X}$ is defined by

$$\mathbb{E}\{\mathcal{X}\} := \sum_{k \in \Omega(\mathcal{X})} k \cdot p(\mathcal{X} = k) \quad (47)$$

- Expectations of functions:

$$\mathbb{E}_x\{f\} := \sum_x p(x) f(x) \quad (48)$$

- **Remark:** In the continuous case we have to replace $\sum$ by $\int$

Expectation (Ctd.)

Rules of expectations:

- The expectation is a **linear** operation:

$$\mathbb{E}\{a\mathcal{X} + b\mathcal{Y}\} = a\mathbb{E}\{\mathcal{X}\} + b\mathbb{E}\{\mathcal{Y}\} \quad (49)$$

- More general: $\mathbb{E}\left\{\sum_{n=1}^N a_n \mathcal{X}_n\right\} = \sum_{n=1}^N a_n \mathbb{E}\{\mathcal{X}_n\}$
- If $\mathcal{X}$ and $\mathcal{Y}$ are independent: $\mathbb{E}\{\mathcal{X}\mathcal{Y}\} = \mathbb{E}\{\mathcal{X}\}\mathbb{E}\{\mathcal{Y}\}$
- The expectation is **monotonous**:

$$\mathcal{X} \leq \mathcal{Y} \implies \mathbb{E}\{\mathcal{X}\} \leq \mathbb{E}\{\mathcal{Y}\} \quad (50)$$

Variance

- The **variance** $\mathbb{V}$ of a random variable $\mathcal{X}$ is defined by

$$\mathbb{V}\{\mathcal{X}\} := \mathbb{E}\{\mathcal{X} - \mathbb{E}\{\mathcal{X}\}\}^2 = \mathbb{E}\{\mathcal{X}^2\} - \mathbb{E}\{\mathcal{X}\}^2 \quad (51)$$

- $\mathbb{V}$ is **not** linear:

$$\mathbb{V}\{a + b\mathcal{X}\} = b^2\mathbb{V}\{\mathcal{X}\} \quad (52)$$

$$\mathbb{V}\{\mathcal{X} + \mathcal{Y}\} = \mathbb{V}\{\mathcal{X}\} + \mathbb{V}\{\mathcal{Y}\} + 2\text{cov}\{\mathcal{X}, \mathcal{Y}\} \quad (53)$$

- BIENAYMÉ's identity:** If $\mathcal{X}$ and $\mathcal{Y}$ are **uncorrelated**, we get:

$$\mathbb{V}\{\mathcal{X} + \mathcal{Y}\} = \mathbb{V}\{\mathcal{X}\} + \mathbb{V}\{\mathcal{Y}\} \quad (54)$$

Covariance

- **Covariances** give a measure of correlation, i. e. how much variables change together

$$\begin{aligned}\text{cov}\{X, Y\} &:= \mathbb{E}\left\{ (X - \mathbb{E}\{X\})(Y - \mathbb{E}\{Y\}) \right\} \\ &= \mathbb{E}\{XY\} - \mathbb{E}\{X\}\mathbb{E}\{Y\}\end{aligned}\tag{55}$$

- The variance $\mathbb{V}$ of a random variable X is a special case:

$$\mathbb{V}\{X\} = \text{cov}\{X, X\}\tag{56}$$



Kullback-Leibler Divergence

- The **Kullback-Leibler (KL) divergence** is a similarity measure between two distributions p and q :

$$\mathbb{KL}(p\|q) := \sum_x p(x) \cdot \log \frac{p(x)}{q(x)} \quad (57)$$

- Some properties:
 - It is not symmetric: $\mathbb{KL}(p\|q) \neq \mathbb{KL}(q\|p)$
 - It is non-negative: $\mathbb{KL}(p\|q) \geq 0$
 - If $\forall x : p(x) = q(x) \implies \mathbb{KL}(p\|q) = 0$

Section: Wrap-Up

Summary
Recommended Literature
Self-Test Questions
Lecture Outlook

Summary

- **Mathematics play a major role in machine learning!**
- **Linear algebra:**
 - You should know what vectors are and what you can do with them (addition, multiplication, transpose, ...)
 - The same applies to matrices
 - You should know the concept of **determinants** and how to **invert matrices**
 - **Eigenvectors** and **eigenvalues** are important tools in machine learning
 - The **eigen-decomposition** plays an import role in many machine learning applications

Summary (Ctd.)

- **Probability theory and statistics:**
 - Random variables are numbers **determined by chance**
 - Probability distributions describe a **probability mass** or **probability density**
 - **Discrete distributions:** BERNOULLI, Binomial, Multinomial
 - **Continuous distribution:** GAUSSIAN distribution
 - GAUSSIANS are important in machine learning and have appealing properties
 - Terms you should know: Joint-, marginal- and conditional distribution, chain rule, probabilistic independence, Bayes' rule
 - You should know what **expectation** and **variance** of distributions are

Recommended Literature

1 Linear algebra:

- [DEISENROTH.2019], chapter 2
- [DEISENROTH.2019], chapter 3
- [DEISENROTH.2019], chapter 4

2 Probability theory and statistics:

- [DEISENROTH.2019], chapter 6

(For free PDF versions, see list in GitHub readme!)



Self-Test Questions

- 1 What is a vector and what is a matrix?
- 2 What is the result of an inner product / outer product?
- 3 How can you invert matrices? Is this always possible?
- 4 What is an eigenvalue problem? How can you compute eigenvectors and eigenvalues?
- 5 What are random variables and probability distributions?
- 6 Why is the GAUSSian distribution so important?
- 7 What is BAYES' rule? Explain its components!

What's next...?

- **I** Machine Learning Introduction
- II** Optimization Techniques
- III** Bayesian Decision Theory
- IV** Non-parametric Density Estimation
- V** Probabilistic Graphical Models
- VI** Linear Regression
- VII** Logistic Regression
- VIII** Deep Learning
- IX** Evaluation
- X** Decision Trees
- XI** Support Vector Machines
- XII** Clustering
- XIII** Principal Component Analysis
- XIV** Reinforcement Learning
- XV** Advanced Regression

Thank you very much for the attention!

***** Artificial Intelligence and Machine Learning *****

Topic: Mathematics Refresher

Term: Summer term 2026

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Do you have any questions?